

Mapping Brain Structure to Function

BrainFacts 3D Brain · Society for Neuroscience

Senior Secondary (Years 11-12)

~30 minutes

Science · Psychology

NOTICE — AI-generated worksheet

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Simulation: <https://www.brainfacts.org/3d-brain>

Curriculum links: brain function · memory · perception · sensation

Learning intentions

- I can identify key brain regions and describe their primary functions.
- I can link specific brain structures to psychological processes such as memory, perception and sensation.
- I can explain how damage to a specific brain region might affect behaviour or cognition.

Getting started

Open the BrainFacts 3D Brain model. Rotate the brain to get a sense of its overall structure, then click through the regions relevant to today's topic (for example, memory, vision or emotion) to read each region's name, location and function before working through the tasks below.

Tasks

1. Locate the hippocampus. What function is it primarily associated with?

2. Locate the occipital lobe. What sense does this region primarily process?

3. Locate the amygdala. What role does it play, based on the information in the model?

4. Click through four brain regions relevant to today's topic and record each one's primary function.

Brain region	Primary function

5. Explain how damage to the hippocampus (for example, from injury or disease) might affect a person's day-to-day life, based on its function.

6. Choose two regions you explored and explain how they might work together during a single everyday task (for example, recognising a friend's face and remembering their name).

7. Locate a region involved in decision-making or planning (such as the prefrontal cortex, if included). How might damage here affect someone's behaviour?

8. Explain why psychologists use case studies of people with specific brain injuries to understand brain function, connecting this to at least one region you explored in the model.

Extension (optional)

Choose a psychological process not directly covered above (such as language, emotion regulation or motor control). Use the 3D Brain model to identify a relevant region, and explain what evidence (case studies, brain imaging, or lesion studies) might have been used to establish that link.
